

MCS-1: Probability and Statistics (MA6.302)

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Lecture 7

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- The conditional PMF of a random variable X , conditioned on a particular event A with $P(A) > 0$, is defined by:

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- But, A can be written as a disjoint union of the events $\{X = x\} \cap A$ for all numerical values X takes.
- Total probability rule gives: $P(A) = \sum_x P(\{X = x\} \cap A)$,
 $P(X = x|A) = 1$.

- Conditional expectation of random variable X conditioned on event A is written as $\mathbb{E}[X|A]$

$$\mathbb{E}[X|A] = \sum_x xP(X = x|A)$$

Conditional Expectation of Geometric Random Variable

Say $X \sim \text{geometric}(p)$. What is $\mathbb{E}[X|X > 1]$?

- Remember $P(X = k|X > 1) = P(X = k - 1)$

So,

$$\begin{aligned}\mathbb{E}[X|X > 1] &= \sum_{k=2}^{\infty} kP(X = k|X > 1) \\&= \sum_{k=2}^{\infty} kP(X = k - 1) \\&= \sum_{k=2}^{\infty} (k - 1)P(X = k - 1) + \sum_{k=2}^{\infty} P(X = k - 1) \\&= \sum_{k=1}^{\infty} kP(X = k) + \sum_{k=1}^{\infty} P(X = k) \\&= \mathbb{E}[X] + 1\end{aligned}$$

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- Move things around: **$\mathbb{E}[X] = \frac{1}{p}$** .

Variance of Geometric Random Variable

$$\bullet \mathbb{E}[X^2] = \underbrace{\mathbb{E}[X^2|X=1]P(X=1)}_{1 \times p} + \underbrace{\mathbb{E}[X^2|X>1]P(X>1)}_{1-p}$$

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- **Substitute $i = k - 1$ to get:**
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- So, plugging everything in, we have:
 $\mathbb{E}[X^2] = p + \mathbb{E}[(1+X)^2](1-p) = p + (\mathbb{E}[X^2] + 2(p+1)) \times (1-p).$

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- Solving, we get $\mathbb{E}[X^2] = \frac{2}{p^2} - \frac{1}{p}.$
- So, $\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1-p}{p^2}.$